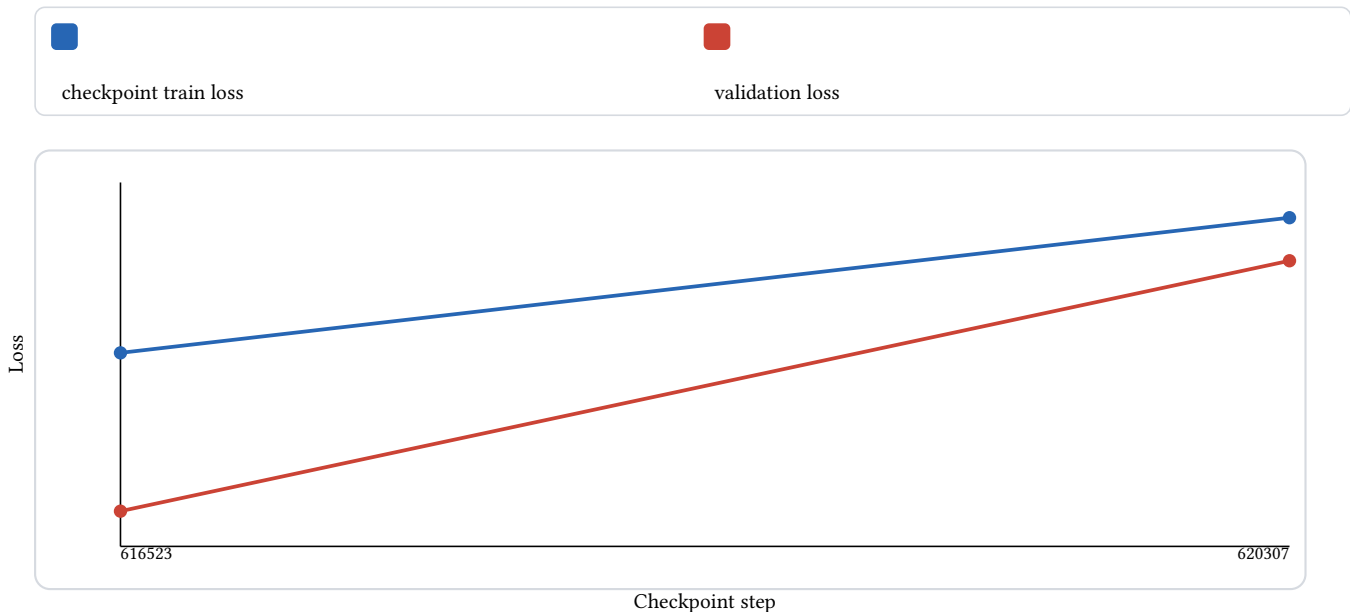


Long ctx2048: 1% ANS tail hurts

38.74M decoder + XSA, exact 20x token budget, 40% curriculum warmup, final comparison on a fresh 1024-example ctx2048 validation set.

Loss overlay



The only change between the two points is the final 1% answer-only tail.

- step 616523: checkpoint right before ANS
- step 620307: final checkpoint after the ANS tail

Accuracy

- pre-ANS checkpoint: $910 / 1024 = 88.87\%$
- final checkpoint after ANS: $764 / 1024 = 74.61\%$
- delta: -14.26 accuracy points

Finding

- The final 1% ANS phase makes both metrics worse: train loss rises, validation loss rises, and exact-match accuracy drops sharply.
- For future Long runs, do not add an ANS tail by default.
- The safer default is to stop at the LM checkpoint before ANS, or only revisit ANS with a much smaller tail and explicit validation gating.

Artifacts

- config: GIANT/v3/Configs/Training/Long/long_40m_ll_ctx2048_chinchilla20x_warm40_5090.yml
- evals: s3://giant-data/GIANT/single-gpu/logs/long/ctx2048_chinchilla20x_warm40_5090/pre_ans_val1024.json
- evals: s3://giant-data/GIANT/single-gpu/logs/long/ctx2048_chinchilla20x_warm40_5090/final_val1024.json
- compare: s3://giant-data/GIANT/single-gpu/logs/long/ctx2048_chinchilla20x_warm40_5090/pre_vs_final_val1024_compare.json